

PPO versus Rule-Based Soft RAN Slicing: Benefits, Limits, and Guarded Acceptance

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Abstract—This paper asks when proximal policy optimization (PPO) provides sufficient benefit to replace deterministic rule-based soft RAN slicing. In our earlier simulations under normal traffic, PPO ties eMBB throughput and reduces mMTC service, fairness, efficiency, and composite reward; a guarded acceptance criterion correctly retains the rule controller. A subsequent exploratory stress test introduces a configurable, held-out offered-load disturbance. Under a severe five-step mMTC access storm, PPO reduces cumulative service deficit from 0.4918 to 0.0052 and raises the minimum observed mMTC service from 0.7456 to 0.9957, while cumulative reward is 1.25% lower and remains within a disclosed 2.5% non-inferiority tolerance. The benefit repeats across five event timings but does not appear for the tested eMBB or URLLC disturbances. Thus PPO is not generally superior; its observed value is conditional resilience under extreme modeled mMTC turbulence. All results are synthetic, use one evaluation seed, and do not measure packet latency or carrier reliability.

Index Terms—RAN slicing, proximal policy optimization, rule-based control, safety, reproducibility

I. INTRODUCTION

Soft RAN slicing changes resource shares as demand changes, potentially reducing the stranded capacity of fixed partitions. The engineering question is not whether PPO can execute, but whether its additional complexity helps under conditions that defeat a transparent rule. This paper therefore evaluates both normal operation and explicit traffic disturbances under matched inputs, actions, safety constraints, and metrics.

Here, *two timescales* has a simple meaning. At the slower timescale, a planner configures the ten-site radio scenario, traffic assumptions, and protected allocation envelope. At the faster timescale, the controller repeatedly decides whether to hold the current slice allocation or transfer 5% between two slices. Thus, slow configuration defines the feasible operating region, while fast control reacts inside it.

The evaluation is *traceable*: the artifact version, state variables, actions, seeds, reward, safety checks, comparison window, and accept/reject rule are stated alongside the result. Traceability here means that an engineering reviewer can follow a result back to its configuration and computation. It is not a claim that the neural network is explainable through feature attribution.

This paper makes three contributions: (i) an auditable seven-input, seven-action formulation for safe soft slicing; (ii) a matched, outcome-neutral acceptance criterion that retains the rule under normal traffic; and (iii) an exploratory disturbance harness that identifies a narrow mMTC resilience benefit and the accompanying reward tradeoff. Unlike an earlier dashboard draft, no result from a separate 13-input demonstration is used.

PPO's clipped objective limits large policy updates [3], but that property does not establish superiority to operational rules. This is an evaluation study, not a new PPO algorithm: both controllers share observations, feasible actions, exogenous inputs, and a criterion fixed before each reported decision. "Soft slicing" denotes transferable shares with protected envelopes, not hard isolation or a complete standards-compliant network slice.

A. Why mMTC Turbulence Is Different

The three service classes create different allocation stresses; their event frequency is deployment-specific, so this paper does not claim that eMBB or URLLC bursts are universally less common. eMBB commonly stresses sustained traffic volume, while URLLC stresses immediate service under strict timing requirements. Massive machine-type communication instead combines individually small transmissions with very high connection density; IMT-2020 identifies up to 10^6 devices/km² as the mMTC connection-density target [4]. 3GPP consequently studied specific RAN overload controls for machine-type communications [5].

An mMTC access storm occurs when many normally quiet devices attempt access within a short interval. Triggers can include synchronized reporting, a common alarm, restoration after a power or coverage outage, firmware-driven reconnects, and aligned timers. Collisions can provoke backoff and retransmission, so attempts may grow faster than the device count; random access is a recognized cellular-IoT bottleneck [6]. This interpretation is compatible with, but not certified by, the standards context: ITU-R defines the massive-connection requirement, while 3GPP TR 37.868 studies MTC access-network overload and controls such as access barring [4], [5]. "Storm" is our shorthand for a synchronized offered-load disturbance; neither reference standardizes the paper's 400% magnitude. The artifact models aggregate service envelopes rather than the complete 3GPP random-access procedure, so the experiment is standards-motivated, not a conformance test.

II. SYSTEM AND CONTROL MODEL

A. Multicell Scenario and Metric Scope

The browser-based model represents ten sites using user-entered site coordinates, radio configuration, and service populations. eMBB traffic may aggregate FDD and TDD capacity; URLLC and mMTC use a single selected carrier. Geometry determines unique coverage and affects modeled mMTC density, access, and retries. The present evaluation concerns allocation

TABLE I

PPO STATE USED IN THE EARLIER SIMULATIONS (SEVEN INPUTS).

Inputs	Meaning
1–3	eMBB, URLLC, and mIoT normalized load
4	Modeled URLLC service
5	eMBB throughput index
6	Jain fairness across modeled slice service
7	Controllable efficiency

TABLE II

CATEGORICAL ACTIONS USED IN THE SIMULATIONS.

a	Action	a	Action
0	Hold	4	mIoT $\rightarrow$ eMBB
1	eMBB $\rightarrow$ URLLC	5	URLLC $\rightarrow$ mIoT
2	eMBB $\rightarrow$ mIoT	6	mIoT $\rightarrow$ URLLC
3	URLLC $\rightarrow$ eMBB		

logic, not a calibrated propagation or packet-level network model.

The default allocation is $(x_e, x_u, x_m) = (0.77, 0.03, 0.20)$ for eMBB, URLLC, and mIoT, with $x_e + x_u + x_m = 1$. The model reports normalized eMBB throughput, modeled mIoT service, Jain fairness, controllable efficiency, and a composite reward. Real URLLC latency is explicitly marked *unverifiable*. It is excluded from the state, reward, policy comparison, and pass/fail decision. The safety layer protects only the modeled minimum URLLC resource envelope; it makes no latency or reliability claim.

B. Seven-Input State

At decision step t , the policy receives

$$s_t = [\ell_e, \ell_u, \ell_m, q_u, T_e, J, E]_t, \quad (1)$$

where the first three terms are normalized service loads, q_u is modeled URLLC service, T_e is the eMBB throughput index, J is Jain fairness, and E is controllable efficiency. Table I gives the semantic grouping. These are model-derived indices rather than measured carrier KPIs.

C. Seven Bounded Actions

The categorical action space is deliberately small. Action 0 holds the shares; the other actions move 0.05 from one slice to another (Table II). This makes every policy output directly inspectable and prevents an unconstrained neural output from becoming a resource allocation.

Before sampling, the controller simulates all seven post-action allocations. Unsafe actions receive zero probability. If no local action is safe, the implementation searches a 0.5-percentage-point allocation grid for a feasible simplex projection; if none exists, control halts instead of emitting an unsafe allocation.

Masking separates preference from permission: PPO ranks actions, while deterministic logic decides which may execute. The rule policy uses the same feasible set, so safety handling does not selectively favor PPO.

III. PPO AND RULE-BASED COMPARATOR

The PPO actor is $7 \rightarrow 64 \rightarrow 64 \rightarrow 7$, with two tanh hidden layers and a categorical softmax output. The critic is $7 \rightarrow 64 \rightarrow 64 \rightarrow 1$. The clipped PPO objective is [3]

$$L^{\text{clip}} = \mathbb{E}_t \left[\min \left(r_t \hat{A}_t, \text{clip}(r_t, 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right) \right]. \quad (2)$$

The reference run uses eight iterations, rollout length 64, four epochs per iteration, $\gamma = 0.97$, generalized-advantage parameter $\lambda = 0.95$, learning rate 0.035, clip $\epsilon = 0.2$, and temperature 0.8. This produces 2,048 sample updates. The actor initialization seed is 99173; rollout i uses seed $12345 + 137i$; matched evaluation uses seed 24680.

The deterministic rule policy is the operational baseline. It applies explicit demand and safety logic to the same state and uses the same bounded transfers. It is not trained, and it is retained unless PPO passes the prespecified gate. Fixed hard slicing is excluded from the numerical comparison because this study evaluates two alternative controllers—PPO and deterministic rule-based control—for the same soft-slicing action space.

At each step, the rule compares normalized slice pressure with the current allocation and chooses a bounded transfer toward the most relevant deficit, subject to the protected envelope. If no permitted transfer improves its deterministic priority, it holds. This is an appropriate comparator for the artifact because it is transparent, reacts to demand, and operates through exactly the same physical actions as PPO. Comparing against a fixed partition alone would not determine whether learning adds value beyond ordinary adaptive control.

For clarity, the notation used in the equations, result table, and figure is fixed throughout the paper: T_e denotes the normalized eMBB throughput index; S_m denotes modeled mIoT service; J denotes Jain's fairness index across modeled slice service; E denotes controllable resource efficiency; and R denotes the dimensionless composite reward. These symbols are performance indices, not percentages. The composite reward is

$$R = 0.40 \text{ clip}(T_e) + 0.25J + 0.15 \text{ clip}(S_m) + 0.20 \text{ clip}(E) - P, \quad (3)$$

The penalty is

$$P = 0.30[T_e^{\min} - T_e]_+ + 0.25[S_m^{\min} - S_m]_+. \quad (4)$$

Reward is a dimensionless ranking objective, not throughput, latency, or reliability. Its weights make the preferred tradeoff explicit and auditable.

IV. MATCHED EVALUATION AND DECISION GATE

Both controllers replay the same 120-step evaluation trace. The summary uses the final 25%, i.e., the last 30 steps, to compare steady-tail behavior rather than mixing early transients with later decisions. PPO is published as accepted only if at least one controllable KPI improves and none degrades. The five gate variables are eMBB throughput, mIoT service,

fairness, controllable efficiency, and composite reward. Equality on every metric is not an improvement; any degradation rejects PPO and retains the rule policy.

This strict gate separates *training demonstrability* from *policy benefit*. Optimization diagnostics can show that gradients and clipping executed, but only matched held-out metrics determine whether the trained policy is preferable.

The evaluation tests a directional engineering hypothesis: after training, PPO should improve at least one controllable metric without sacrificing another relative to the incumbent rule. The null operational decision is to retain the rule. This framing differs from a conventional significance test; our earlier simulations use one reference trace and do not estimate a sampling distribution. The gate is instead a conservative deployment screen. A future statistical study would repeat the paired experiment across independently generated traces and seeds before making a general performance claim.

The replay freezes the ten-site configuration and initial 77/3/20 allocation, clears training accumulators, and gives both controllers the identical evaluation-seed sequence. Each action passes through the same safety logic, and the tail window is fixed before inspection. The audit record stores version, inputs, architecture, seeds, reward, safety floors, window, raw summaries, and decision. Modeled URLLC service remains distinct from unmeasured packet latency.

V. RESULTS

Our earlier simulations under normal traffic give rule/PPO values of 1.2500/1.2500 for eMBB throughput, 1.1000/1.0865 for mIoT service, 0.6362/0.5634 for fairness, 1.1975/1.1928 for efficiency, and 0.9090/0.8908 for reward. PPO therefore ties only eMBB and is rejected.

The largest difference is the fairness reduction of 0.0728. The reward reduction of 0.0182 is consistent with (3), because lower fairness and mIoT service offset the eMBB tie.

The fairness and reward differences are approximately -11.4% and -2.0% of the rule values. They are descriptive ratios, not confidence intervals. Reward is the weighted objective in (3), not an SLA-compliance percentage; component metrics remain necessary to expose tradeoffs hidden by aggregation.

A. Subsequent Exploratory Disturbance Result

The subsequent internal artifact [8] applies a held-out offered-load disturbance only during matched evaluation; it is absent from PPO training. The user selects slice, percentage increase, start step, and duration. Cumulative service deficit is

$$Q = \sum_{t \in \mathcal{W}} [1 - S_t]_+, \quad (5)$$

over the disturbance plus ten recovery steps. At each step, service target 1.0 is subtracted from the achieved service only when the target is missed; therefore $Q = 0$ is best and a larger Q means more accumulated under-service. The minimum service is simply the lowest single service value observed in the window, so higher is better. Mean reward is the average weighted score across all modeled objectives, so

higher is better. A stress-specific acceptance criterion requires at least 10% lower Q , no lower minimum service, and reward within a disclosed 2.5% non-inferiority tolerance.

A predefined exploratory sweep covered three slices and 50, 100, 200, and 400% disturbances. No benefit appeared for the tested eMBB or URLLC events. For a five-step 400% mIoT increase beginning at step 60, PPO reduces accumulated under-service Q from 0.4918 to 0.0052 and raises minimum mIoT service from 0.7456 to 0.9957. Mean reward is 0.9162 for the rule and 0.9047 for PPO (cumulative values 13.7425 and 13.5701), a 1.25% reduction. Figure 1 shows that PPO protects mIoT service during the storm but pays a small overall-objective cost.

Moving the same event to steps 25, 40, 60, 80, and 100 gives rule shortfalls of 0.6135, 0.4315, 0.4918, 0.3423, and 0.4364; PPO shortfalls are 0, 0.0089, 0.0052, 0.0008, and 0.0080. Reward differences range from $+0.49\%$ to -2.00% . This timing check is encouraging but not confirmatory: it uses the same traffic generator, one evaluation seed, and an event magnitude selected after the exploratory sweep.

VI. TRACEABILITY, LIMITATIONS, AND VALIDITY

The cited v15.4 and v16 software artifacts are internal and unpublished [7], [8]. IEEE style permits citing such material when its status is clear, although it is not independently retrievable like an archival publication. The citations identify the implementations that fix the state, actions, seeds, reward, windows, disturbance, and acceptance criteria; the paper supplies the necessary details rather than relying on the citations alone.

The results are synthetic and configuration-specific. They do not establish operator-network performance, packet latency, block-error rate, availability, or carrier-grade reliability. The single evaluation seed does not quantify training variance. The v16 magnitude and 2.5% reward tolerance were selected during exploratory analysis and therefore require validation on frozen, independent tests. The reward weights encode an engineering preference, and changing them could change the ranking. A seven-action controller is inspectable but coarse, while the safety mask protects modeled allocation constraints rather than all radio conditions.

Construct validity is limited because all KPIs are modeled proxies. Matched replay improves internal validity, but conclusions still depend on seed, training budget, and hyperparameters. External validity is limited by the synthetic ten-site configuration; results cannot be generalized to another topology, scheduler, or traffic mixture without rerunning the protocol.

Future work should freeze the v16 disturbance matrix and 2.5% tolerance before testing independent traffic generators and seeds, then report confidence intervals, win/tie/loss rates, and effect sizes. Packet-level experiments should model random access, backoff, retransmissions, scheduler latency, BLER, signaling overhead, and rollback. The reported controller uses a fixed five-percentage-point transfer; a future implementation should expose 0–5% transfer magnitude so

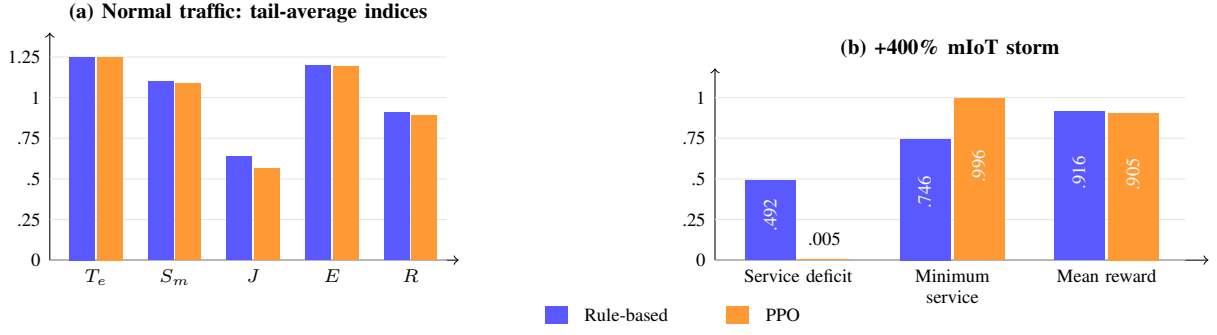


Fig. 1. Matched rule-based and PPO results. In (a), higher is better. In (b), service deficit is accumulated failure to reach target 1.0 (lower is better); minimum service is the lowest value during the event (higher is better); mean reward is the average overall weighted score (higher is better).

uncertain estimates can trigger more conservative changes. Most importantly, the mIoT result should remain conditional unless it survives independent validation.

VII. CONCLUSION

PPO is not generally superior to deterministic rule-based soft slicing. In our earlier simulations under normal traffic, guarded acceptance rejects PPO. In the subsequent severe mIoT storm, PPO nearly eliminates accumulated service deficit and raises minimum service while accepting a small reward reduction; the tested eMBB and URLLC disturbances show no corresponding benefit. The defensible contribution is therefore conditional selection: retain transparent rules for ordinary operation and consider PPO only where repeated, independent evidence shows that resilience justifies its complexity and tradeoffs.

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